

Lab 8 - Splunk SIEM Investigation

Windows Authentication Monitoring and Failed-Logon Detection

Lab objective	Deploy Splunk Enterprise, ingest Windows Security events, investigate authentication activity with SPL, and create a scheduled failed-logon alert and monitoring dashboard.
Splunk platform	Splunk Enterprise 10.4.3 on Ubuntu Server 24.04.5 LTS
Windows endpoint	Windows 11 VM with Splunk Universal Forwarder
Indexer address	192.168.180.136:9997
Primary log source	Windows Security Event Log
Key event IDs	4624 - successful logon; 4625 - failed logon
Exercise type	Authorized controlled lab simulation

Executive Summary

Splunk Enterprise was installed on a dedicated Ubuntu Server VM and configured to receive Windows telemetry on TCP port 9997. A Splunk Universal Forwarder on the Windows 11 endpoint forwarded the Security event log to the main index. SPL searches were used to identify successful and failed authentication events, and a custom detection was created to identify bursts of three or more Event ID 4625 failures. The scheduled alert was successfully triggered during a controlled test, and a dashboard was built to summarize authentication activity.

Outcome: End-to-end Windows authentication monitoring and detection in Splunk was successfully demonstrated.

1. Architecture and Configuration

Component	Role	Configuration
Ubuntu Server VM	Splunk Enterprise indexer/search head	Ubuntu 24.04.5 LTS; Splunk 10.4.3; TCP 8000 (Web), 9997 (receiving)
Windows 11 VM	Log source / monitored endpoint	Splunk Universal Forwarder; Windows Security log enabled
VMware NAT	Lab network	Endpoint and indexer communicated on 192.168.180.0/24

Windows event input

```
[WinEventLog://Security] disabled = 0 index = main renderXml = true
```

The forwarder configuration was verified with Splunk btool, and the forwarder log showed successful connections to the indexer at 192.168.180.136:9997.

2. Authentication Investigation

The Security log was searched for Event IDs 4624 and 4625. Because the XML sourcetype did not automatically expose EventCode in the required search context, SPL rex extraction was used to parse the EventID from raw XML.

```
index=main sourcetype="XmlWinEventLog:security" ("4624" OR "4625") | rex field=_raw "(?\\d+)" | stats count by EventCode
```

Initial controlled test results showed 16 successful logon events (4624) and 3 failed logon events (4625). Additional controlled failures were later generated to test the scheduled alert.

3. Failed-Logon Detection

A detection query was created to identify a burst of three or more failed interactive logons within the search window. The test events were local interactive attempts (LogonType 2) and showed source IP 127.0.0.1, so they were treated as local lab activity rather than evidence of a remote attack.

```
index=main sourcetype="XmlWinEventLog:security" "4625" | rex field=_raw "(?<\\d+)" | rex field=_raw "(?<\\d+)" | rex field=_raw "(?<\\.\\*\\*)" | stats count earliest(_time) as FirstSeen latest(_time) as LastSeen by host IpAddress LogonType | where count >= 3 | convert ctime(FirstSeen) ctime(LastSeen)
```

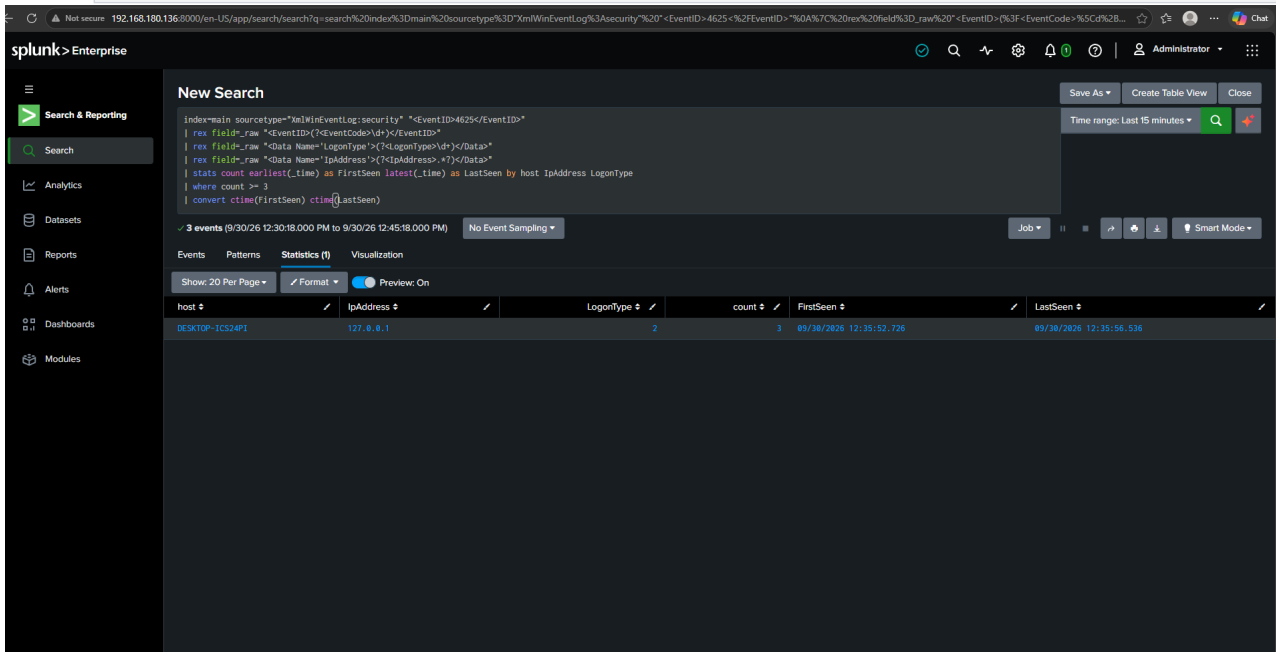


Figure 1. SPL detection summarized three failed Event ID 4625 logons from localhost with LogonType 2.

Detection result

Host	DESKTOP-ICS24PI
Source IP	127.0.0.1 (localhost)
Logon Type	2 - interactive local logon
Failed attempts	3
First seen	2026-09-30 12:35:52.726
Last seen	2026-09-30 12:35:56.536

4. Scheduled Alert and Dashboard

The detection was saved as a scheduled alert named Multiple Failed Windows Logons. It ran every five minutes and triggered when the search returned at least one summarized result. The alert action added the detection to Splunk Triggered Alerts.

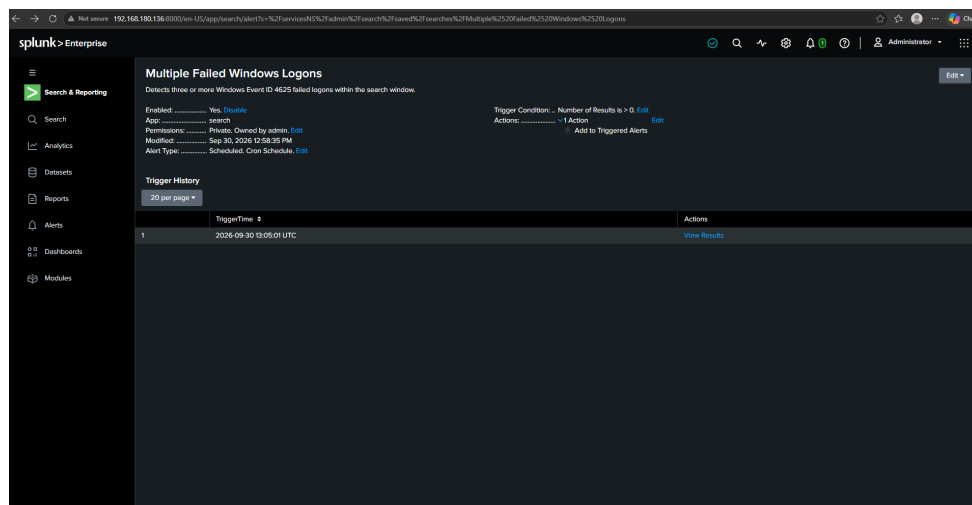


Figure 2. The scheduled failed-logout alert successfully triggered during the controlled test.

A classic dashboard named Windows Authentication Monitoring was created with two panels: a column chart comparing successful and failed logons, and a table summarizing failed-logout activity by host, IP address, and logon type.

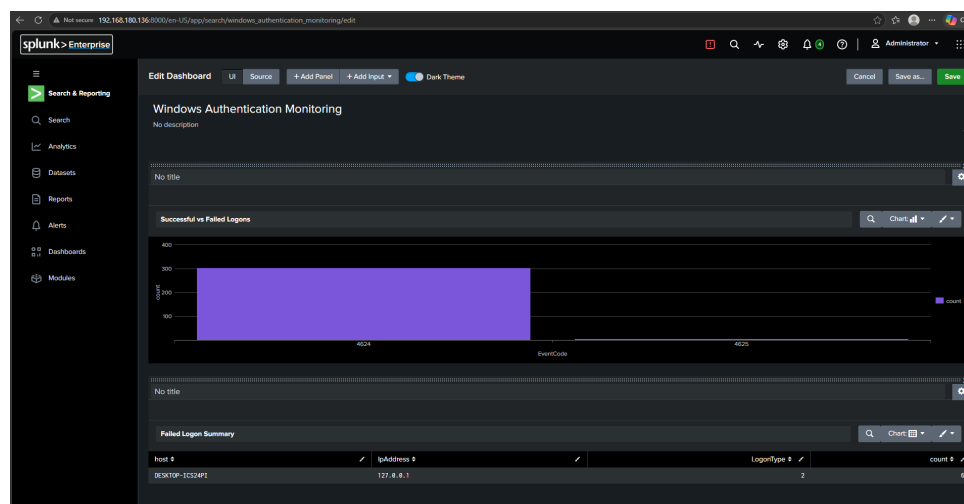


Figure 3. Final Splunk dashboard showing authentication monitoring and failed-logout summary.

5. Analyst Assessment

Classification: Controlled / authorized security test. The evidence demonstrates successful SIEM ingestion, search, correlation, detection engineering, alerting, and dashboard creation. The 4625 events were generated intentionally on the local Windows VM; the localhost address and LogonType 2 do not support a claim of remote brute-force activity.

Skills demonstrated: Splunk Enterprise deployment, Universal Forwarder configuration, Windows Security log ingestion, SPL search and field extraction, Event ID 4624/4625 analysis, scheduled alert creation, trigger validation, and dashboard construction.

Lab status: Completed.